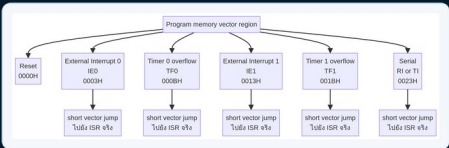


41 Interrupt Mechanism

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Interrupt [1][2]

When CPU
executes
ISR
it temporarily
suspends
the execution of
the program
and
executes
the ISR



EVENT-DRIVEN

Interrupts are generated by external events



STATE-PRESERVING

Interrupts do not affect the state of the processor



VECTORED

Vector table points to the handler



PRIORITY-AWARE

Enable, mask priority interrupts

Interrupt processing flow 3 [1][2][3][4]

PHASE 01

REQUEST

1 Event / flag
Pending

PHASE 02

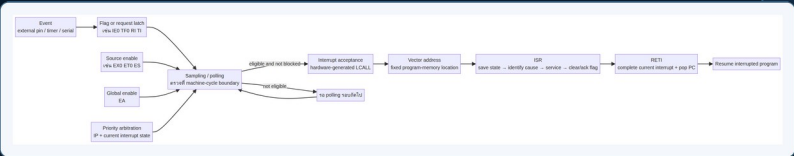
ENTRY

2-4 Qualify → Save → Vector
handler

PHASE 03

SERVICE / RETURN

5-6 ISR / acknowledge → restore
resume



Interrupt State Latency [2][3][4]

01 • QUALIFY

ENABLE / MASK / PRIORITY

control flow

02 • PRESERVE

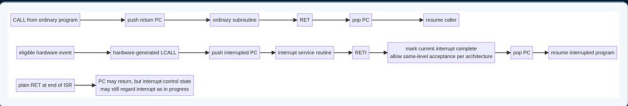
CONTEXT INTEGRITY

Save / Restore ABI

03 • COMPLETE

BOUNDED ISR

Acknowledge latency jitter





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M. A. Mazidi, S. Naimi and S. Naimi. The AVR Microcontroller and Embedded Systems. pp. 375–376, Prentice Hall.

[2] Microchip Technology Inc. (2018).

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[3] Yiu, J. (2010).

The Definitive Guide to the ARM Cortex-M3. 2nd ed. pp. 172–175, Newnes.

[4] Arm Ltd. (2010).

Cortex-M3 Devices Generic User Guide (DUI0552/a). <https://developer.arm.com/documentation/dui0552/a> [10 pp. 2569]